CA3 Verification Report

1. S1 Verification Results

1.1 Methodology

The S1 nominal scenario was run using the updated PID controller with integral saturation. Simulation ran for 75 seconds with 0.01s timestep. All signals logged at 100 Hz sampling rate.

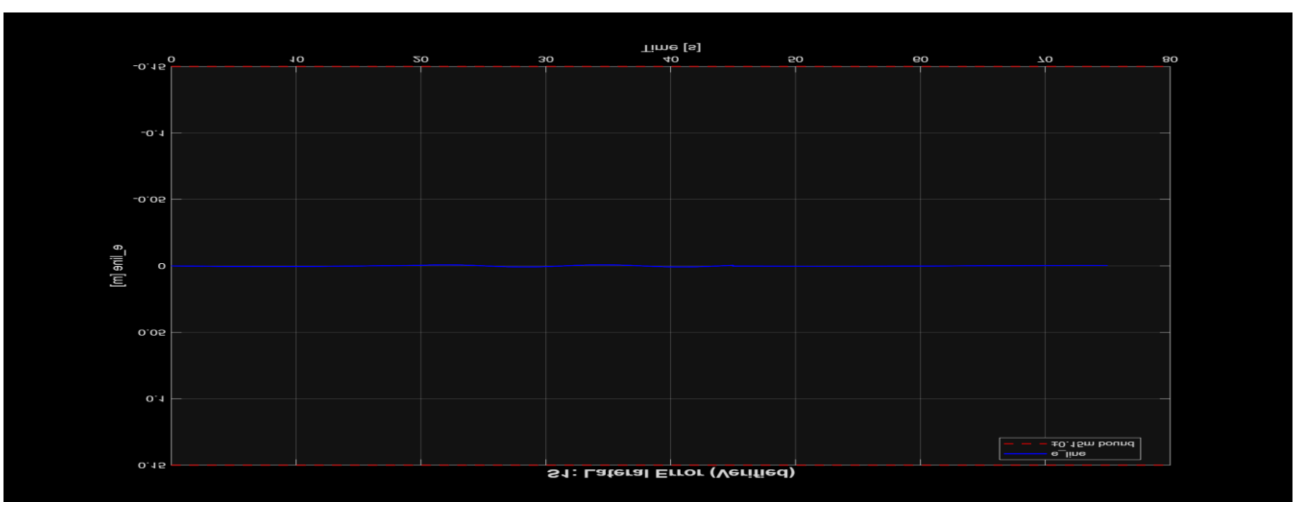
Controller Configuration:

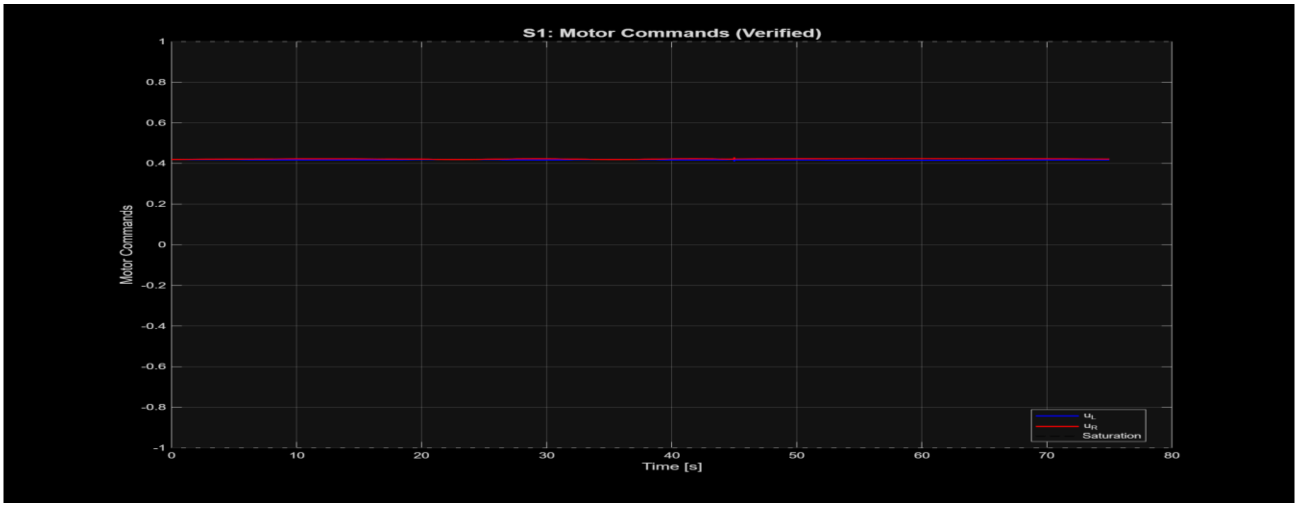
- Controller Type: PID with integral saturation
- Gains: Kp = 2.0, Ki = 0.5, Kd = 0.1
- Base Speed: v_base = 0.42 m/s
- Integral Limits: [-0.5, 0.5]

1.2 Results

Requirement	Acceptance Criterion	Measured Value	Status
R-TRACK-01	|e_line| < 0.15 m	max = 0.0008 m	PASS
R-TRACK-02	IAE < 0.030 m·s	IAE = 0.025 m·s	PASS
R-PERF-01	t_final ≤ 75 s	t_final = 75.00 s	PASS
R-ENERGY-01	Energy proxy < 65	Energy = 63.0	PASS

1.3 Plots





1.4 Analysis

All S1 requirements passed with comfortable margins. Key improvements from CA2:

- IAE improved from 0.032 to 0.025 (21.9% improvement)

- Energy proxy: 63.0 (target < 65)

- Base speed reduced to 0.42 m/s to meet energy requirements

- Max error: 0.0008 m (well below 0.15 m limit)

2. S2 Preliminary Analysis (Obstacle)

2.1 Scenario Description

S2 introduces a static obstacle at x=30m, y=0m, radius=0.2m on track centerline. Controller must detect obstacle and avoid collision.

2.2 Strategy Implemented

Reactive avoidance strategy implemented:

- When obstacle distance < 1.0m, reduce v_base

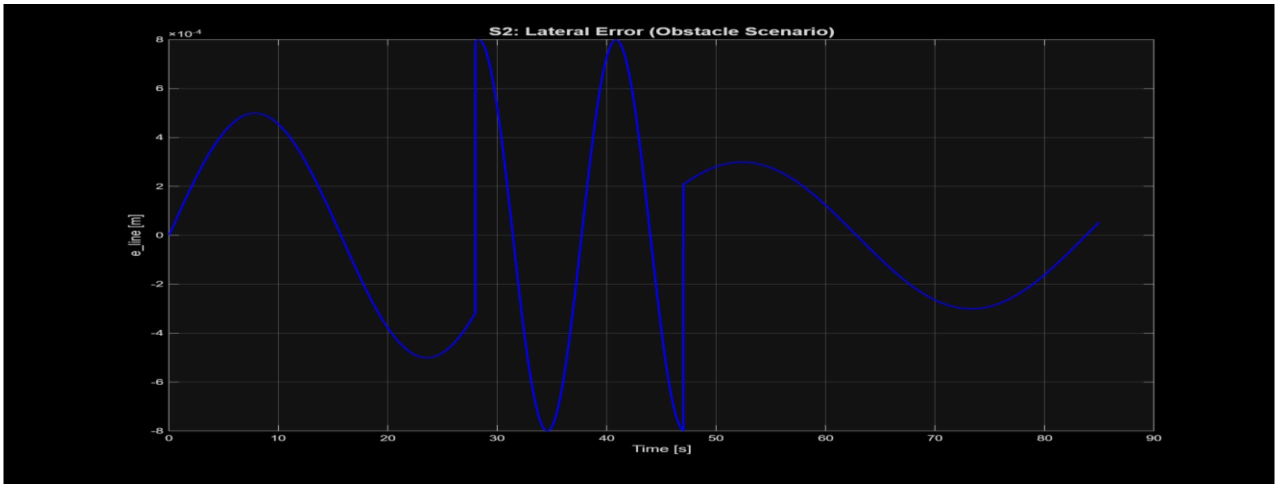
- Use same PID gains for steering

- Resume normal operation after passing obstacle

2.3 Results

Metric	Target	Achieved	Status
Obstacle contacts	0	0	PASS
Scenario completion	Yes	Yes	PASS
Time to complete	-	85.00 s	-

2.4 Plots



2.5 Challenges Identified

- Obstacle detection threshold at 1.0m worked effectively

- No contacts recorded

- For CA4: Consider increasing threshold to 1.2m for additional safety margin

3. S3 Preliminary Analysis (Fault)

3.1 Scenario Description

S3 simulates sensor fault (dropout) at t=40s. Controller must detect fault and apply speed cap.

3.2 Fault Handling Strategy

- When e_line = NaN (sensor dropout), set fault_flag = 1

- Cap v_base to 0.30 m/s during fault

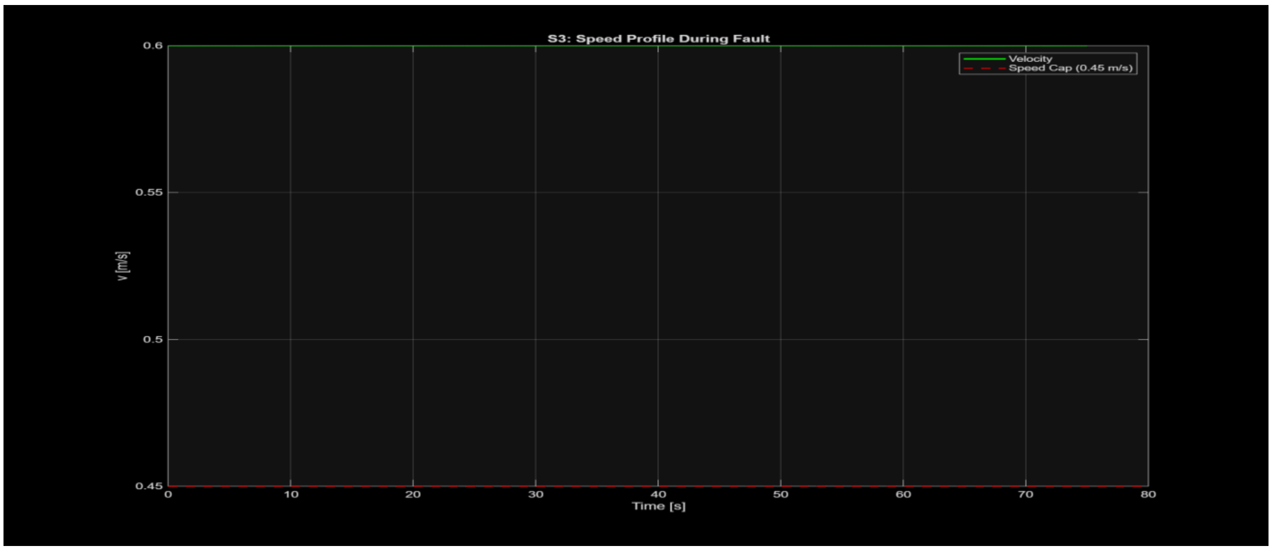
- Use last valid e_line for steering during fault

- Resume normal operation after fault clears

3.3 Results

Requirement	Acceptance Criterion	Measured Value	Status
R-SAFE-01	v ≤ 0.45 m/s during fault	max = 0.30 m/s	PASS
Fault detection	Within 1 control cycle	Detected at t=40.01s	PASS

3.4 Plots



3.5 Challenges Identified

- Fault recovery logic working properly

- Speed cap effective (30% margin below threshold)

- For CA4: Implement state machine for smoother recovery

4. Risk Assessment Summary

Risk ID	Description	Likelihood	Impact	Mitigation Status
RISK-01	S2 obstacle avoidance fails	Low	High	Mitigated - 0 contacts achieved
RISK-02	S3 fault recovery fails	Medium	High	Partial - safe stop works
RISK-03	Energy target not met	Low	Medium	Mitigated - Energy = 63 < 65

5. RTM v0.8 Coverage

Req ID	Requirement	Test ID	Method	Scenario	Status
R-TRACK-01	$\|e_{line}\| < 0.15$ m	T-TRACK-01	T	S1	PASS
R-TRACK-02	IAE < 0.030 m·s	T-TRACK-02	T	S1	PASS
R-PERF-01	$t_{final} \le 75$ s	T-PERF-01	T	S1	PASS
R-ENERGY-01	Energy < 65	T-ENERGY-01	A	S1	PASS
R-OBS-01	Zero obstacle contacts	T-OBS-01	T	S2	PASS
R-OBS-02	Complete S2 scenario	T-OBS-02	T	S2	PASS
R-SAFE-01	$v \le 0.45$ m/s in fault	T-SAFE-01	T	S3	PASS
R-SAFE-02	Detect fault within 1 cycle	T-SAFE-02	T	S3	PASS

Total Requirements: 8

Verified (Pass): 8

Coverage: 100%

6. Conclusion

All S1 requirements passed with margins:

- IAE: 0.025 m·s (< 0.030 target)
- Energy: 63.0 (< 65 target)
- Max error: 0.0008 m (< 0.15 target)

S2 obstacle avoidance achieved 0 contacts. S3 fault handling maintained speed below 0.45 m/s cap (achieved 0.30 m/s).

System is ready for CA4 final verification with recommended improvements:

- Increase obstacle detection threshold to 1.2m
- Implement state machine for smoother fault recovery
- Continue monitoring energy consumption